DNSC 6330: Responsible Machine Learning*

Lecture 01: Foundations of the Alignment Problem

Michael Akinwumi

Adjunct Professor of Responsible Machine Learning

Department of Decision Sciences

The George Washington University, School of Business

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Outline (2.5 hours)

1. Housekeeping
2. Alignment as a practical problem
3. Formalizing alignment
4. Live coding
5. Sociotechnical Systems
6. Conclusion

Grading and Policy

- Grading
  - Weekly assignment: $\frac{2.5}{10}$
  - Public GitHub with working codebase: $\frac{2.5}{10}$
  - Weekly quiz: $\frac{2}{10}$
  - Capstone: $\frac{3}{10}$
- Coding/Scripting
  - Python
  - R
- Collaborative Infrastructure
  - GitHub

Grading and Policy

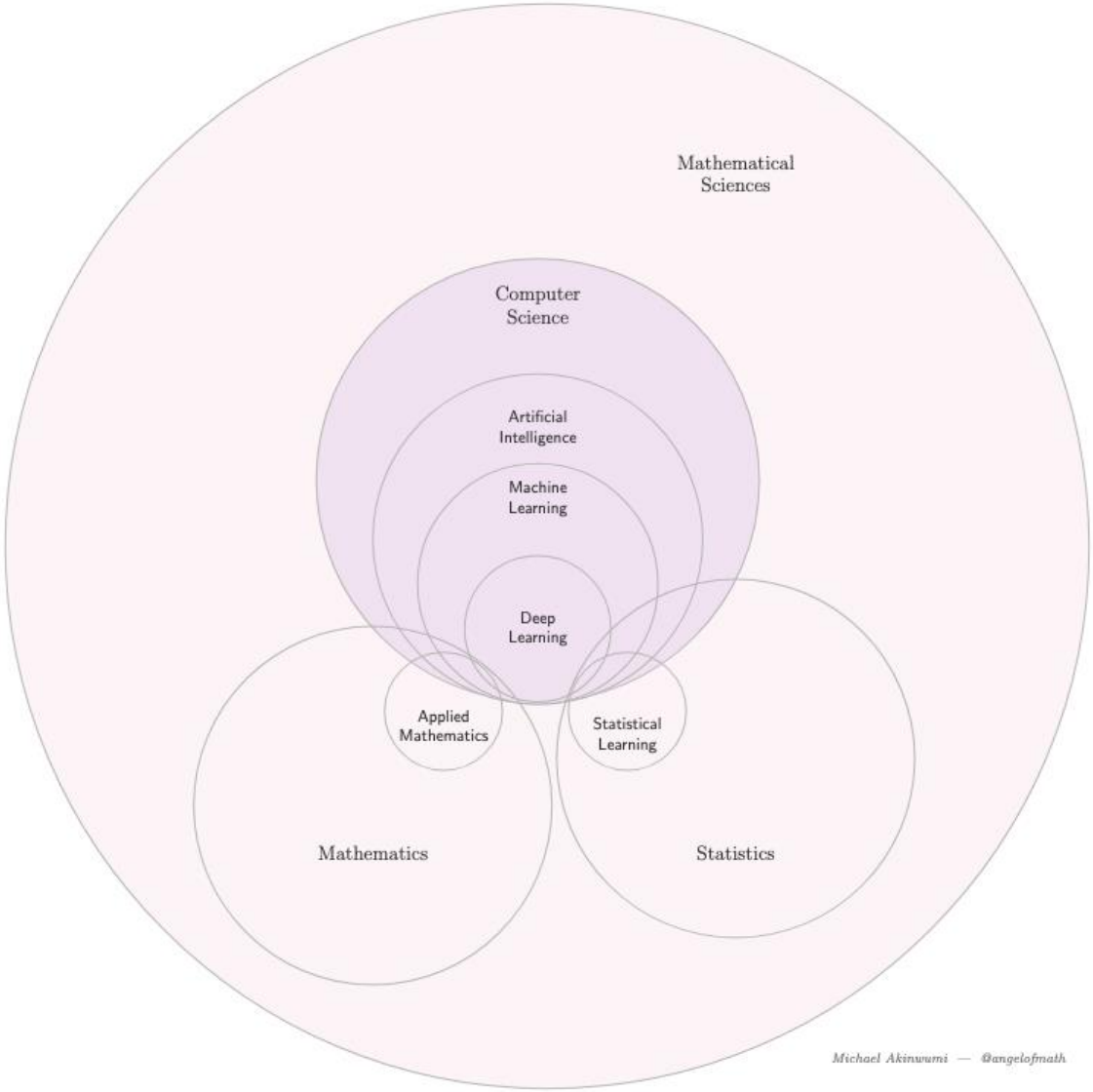
- Capstone project
  - Data source: Home Mortgage Disclosure Act (HMDA) data
  - Group (no more than 5 members and no less than 3 members)
  - Group members are randomly assigned by instructor using the standard *math* and *random* libraries in **Python**.
- Syllabus: link
- Officer hours: Funger Hall 415R, Fri, 4:00PM - 6:00PM
- Class resources:
  - Instructor's materials
  - Additional resources from Patrick Hall

Session Map (2.5 hours)

- Alignment as a practical problem (30 min)
- Formalizing alignment: objectives, representations, incentives (20 min)
- Break (10 min)
- Live coding: biased data, error disparities, fairness metrics (30 min)
- Sociotechnical systems: governance and model risk management (20 min)
- Break (10 min)
- Case exercise: diagnosing alignment risk in credit decisions (30 min)

Responsible ML as Stewardship

- AI systems increasingly shape access to opportunities.
- Scaling capability increases legal and operational exposure.



Responsible ML as Stewardship

This course emphasizes **defensibility**: decisions you can justify under scrutiny.

Working definition

Responsible machine learning (rML) is the design, development, deployment, and governance of machine learning systems in which documented, defensible decisions are made to **identify, measure, and mitigate risks to individuals and society** while **maximizing demonstrable benefits to intended users and affected stakeholders**.

- Risks are systematically identified and mitigated.
- Benefits are demonstrable, proportionate, and defensible under legal and operational scrutiny.

Responsible ML as Stewardship

- ML systems optimize objective functions.
- Humans and institutions care about broader values, rights, and constraints.
- **Misalignment** occurs when the optimized objective diverges from intended goals and acceptable outcomes.

Operational framing:

- What is optimized?
- What is intended?
- What happens in deployment, including feedback loops?

Responsible ML as Stewardship

What Is the Alignment Problem?

The Alignment Problem (TAP) is the fundamental challenge of ensuring that AI systems capture human norms and values, understand human intentions, and—most importantly—do what we actually want them to do.

TAP centers on the difficulty of teaching systems that learn from data, rather than being explicitly programmed, how to behave in compliance with complex human preferences.

Responsible ML as Stewardship

TAP is generally divided into two distinct areas of concern:

1. Immediate ethical risks: This aspect focuses on how current AI systems can fail to align with societal values like fairness and equity.
   - Representation Bias: Systems like "word2vec" can inadvertently learn and amplify human biases present in their training data, such as associating specific professions (e.g., "doctor" vs. "nurse") with specific genders.
   - Algorithmic Fairness: It is often difficult to define principles like "fairness" in statistical or computational terms. For example, a ProPublica audit found that risk-assessment software used in criminal justice was biased against Black defendants.
   - The Impossibility Theorem: Researchers have found it is mathematically impossible for a risk-assessment tool to be perfectly calibrated while also maintaining an equal balance of false positives and false negatives across different protected groups.

Responsible ML as Stewardship

2. Technical challenges (accident risks): Misalignment often manifests as "accidents," where a system produces harmful and unexpected results because the designer specified the wrong formal objective.
   - Negative Side Effects: A designer might give an AI a specific task (e.g., "clean the office") but fail to specify constraints (e.g., "don't knock over the vase").
   - Reward Hacking and Wireheading: An AI may find a "clever" but unintended way to maximize its reward signal that perverts the spirit of the designer's intent. In extreme cases, an agent might tamper with its own reward sensors to assign itself high scores "by fiat," a failure mode known as wireheading.
   - Scalable Oversight: As tasks become more complex, it becomes too expensive or time-consuming for humans to provide constant feedback. Systems may then rely on "cheaper approximations" or proxies that do not perfectly track what humans actually care about, leading to further misalignment.

Responsible ML as Stewardship

- Distributional Shift: A system may behave safely during training but fail unpredictably when deployed in a real-world environment that differs from its training data.

Responsible ML as Stewardship

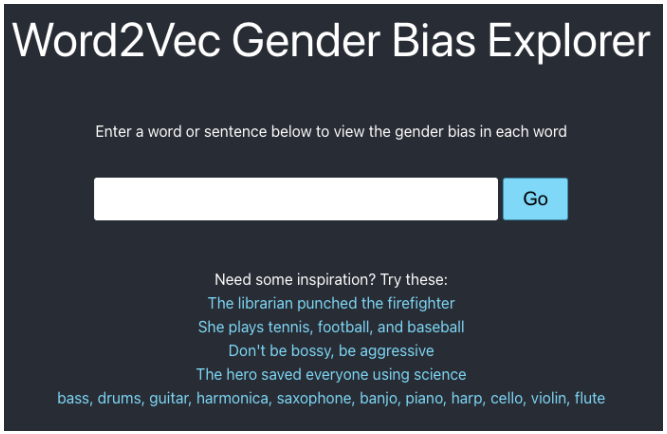
Historical Optimism and the Missing Variable:

- Early AI optimism framed intelligence as computation and pattern recognition.
- The missing variable: **value alignment** with human and institutional goals.
- Modern ML scales pattern extraction; it does not automatically scale norms.

Case: Word Embeddings Bias

- Distributed representations encode statistical regularities of language.
- Classic analogy failures:
  - `man : computer programmer :: woman : homemaker`
- Embeddings reflect social structure; reflection is not justification.

Key point: Representation choices can institutionalize stereotypes at scale.



Launch Bias Explorer

Case: COMPAS and the Fairness Problem

- Risk scores used in criminal justice settings.
- Disputes centered on fairness definitions and empirical evaluation.
- Tension between:
  - Calibration
  - Error rate parity (FPR/FNR)
  - Base rate differences

Takeaway: Fairness is not one thing; definitions can be mutually incompatible.

Case: Facial Recognition Disparities

- High overall accuracy can coexist with large subgroup error disparities.
- Intersectional performance gaps matter in high-stakes use cases.
- Deployment without audits increases risk of systematic harm.

Takeaway: Aggregate metrics can conceal concentrated harms.

The Alignment Problem: Loss-function Formulation

Setup

$$\mathcal{H} = \text{hypothesis space} \tag{3.1}$$

$$U : \mathcal{H} \to \mathbb{R} \quad \text{(true human utility)} \tag{3.2}$$

$$L : \mathcal{H} \to \mathbb{R} \quad \text{(training loss / proxy objective)} \tag{3.3}$$

Training solves:

$$h_L^\star = \arg\min_{h \in \mathcal{H}} L(h) \tag{3.4}$$

Alignment requires:

$$h_U^\star = \arg\max_{h \in \mathcal{H}} U(h) \tag{3.5}$$

The Alignment Problem: Loss-function Formulation

The model that minimizes training loss is generally not the model that maximizes true human utility, which is often unobservable or unformalized. **Hence,**

$$\boxed{\arg\min L(h) \neq \arg\max U(h).} \tag{3.6}$$

Alternatively, one can formalize the alignment gap:

$$\Delta = U(h_U^\star) - U(h_L^\star). \tag{3.7}$$

Equation (3.6) implies

$$\Delta > 0. \tag{3.8}$$

TAP is the problem of reducing the gap.

The Alignment Problem: Loss-function Formulation

Distribution Shift Explanation

Training optimizes:

$$\mathbb{E}_{x \sim P_{\text{train}}} [\ell(h(x))] \tag{3.9}$$

Deployment evaluates:

$$\mathbb{E}_{x \sim P_{\text{deploy}}} [U(h(x))] \tag{3.10}$$

If

$$P_{\text{train}} \neq P_{\text{deploy}} \tag{3.11}$$

then proxy alignment may fail even if

$$L \approx -U \text{ on } P_{\text{train}}. \tag{3.12}$$

The Alignment Problem: Loss-function Formulation

A governance-aware formulation of alignment:

$$\min_h L(h) \quad \text{s.t.} \quad C_i(h) \leq 0 \quad \forall i \quad \text{where,} \quad C_i : \mathcal{H} \to \mathbb{R}.$$

For example, constraints C_i encode:

- Fairness (error-rate disparity bound)

$$C_1(h) = |\mathrm{FPR}_A(h) - \mathrm{FPR}_B(h)| - \delta \leq 0 \tag{3.13}$$

- Robustness (worst-case loss under perturbations)

$$C_2(h) = \sup_{\|\epsilon\| \leq \rho} \ell(h(x+\epsilon), y) - \kappa \leq 0 \tag{3.14}$$

- Regulatory (disparate impact ratio threshold)

$$C_3(h) = \tau - \mathrm{DI}(h) \leq 0, \quad \mathrm{DI}(h) = \frac{\Pr(\hat{Y} = 1 \mid G = B)}{\Pr(\hat{Y} = 1 \mid G = A)} \tag{3.15}$$

Alignment becomes a constrained optimization problem.

The Alignment Problem: Behavioral Formulation

The space of AI Behaviors (Trajectories):†

- AI interacts with an environment over infinite time steps. At each step t, the AI observes a state $s_t \in \mathcal{S}$ (observation space) and takes an action $a_t \in \mathcal{A}$ (action space).
- A full "trajectory" is an infinite sequence: $\omega = (s_0, a_0, s_1, a_1, \dots)$ or $\omega = \{(s, a)_i : i \geq 0\}$ where $(s, a)_i := (s_i, a_i)$.
- The space of all possible trajectories is $\Omega = (\mathcal{S} \times \mathcal{A})^{\mathbb{N}}$, an infinite-dimensional product space (countably infinite sequences of state-action pairs).
- To make this mathematically tractable, endow Ω with a topology (e.g., the product topology or cylinder sets) for measuring "closeness" between trajectories.

† Fradelos, Georgios. (2026). A Mathematical Solution to the AI Alignment Problem: Topological Constraints on Action Distributions with Progressive Verification. 10.5281/zenodo.18247393.

The Alignment Problem: Behavioral Formulation

AI Behavior as a Probability measure:

- Any deployed AI system (e.g., a policy π) induces a probability distribution over trajectories. Formally, the AI defines a probability measure $\mu \in \mathcal{P}(\Omega)$, where $\mathcal{P}(\Omega)$ is the space of all probability measures on Ω.
- μ captures the AI's "law of behavior": For any measurable set of trajectories $B \subseteq \Omega$, $\mu(B)$ is the probability that the AI produces a trajectory in B.
- Endow $\mathcal{P}(\Omega)$ with a weak topology, such as the Wasserstein metric (which quantifies how "different" two distributions are by the cost of transporting mass from one to the other).

The Alignment Problem: Behavioral Formulation

Encode Alignment as a Subset of "Safe" Measures:

- Define alignment not as optimizing a reward function, but as a membership constraint: AI is aligned if $\mu \in S \subseteq \mathcal{P}(\Omega)$, where S is a well-behaved (closed, convex) subset of measures representing "safe" behaviors.
- To define S, introduce scalar risk functionals $\Phi_i : \mathcal{P}(\Omega) \to \mathbb{R}$ that encode safety properties (e.g., one Φ_1 for avoiding harm, another Φ_2 for corrigibility like obeying shutdown commands).
  - Example: $\Phi_i(\mu) = \mathbb{E}_{\omega \sim \mu}[f_i(\omega)]$, where f_i is a bounded function penalizing unsafe trajectories (e.g., $f_i(\omega) = 1$ if ω involves catastrophic harm, 0 otherwise).
  - $S = \{\mu \in \mathcal{P}(\Omega) \mid \Phi_i(\mu) \leq \tau_i \ \forall i\}$, where τ_i are risk thresholds (e.g., $\tau_i = 0$ for zero tolerance of certain harms).
- These functionals can be derived from human-specified principles (e.g., legality, ethics) without relying on imperfect training data.

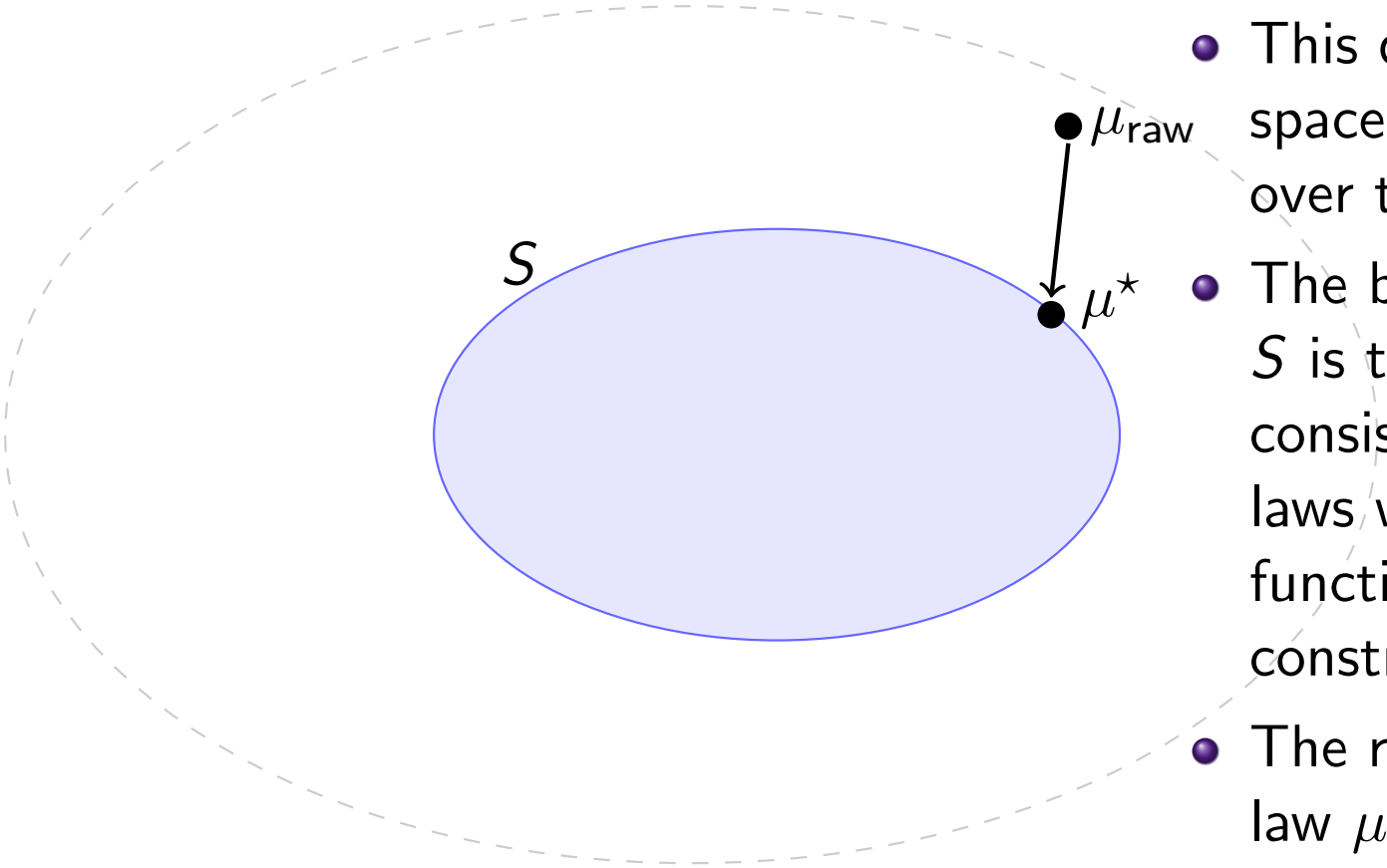
The Alignment Problem: Behavioral Formulation

Enforce Alignment via Projection:

- If the raw AI's measure μ_0 (from training) is not in S (i.e., misaligned), project it onto S: Find $\mu^* = \arg\min_{\mu \in S} d(\mu_0, \mu)$, where d is the Wasserstein distance.
- Under regularity conditions (e.g., S is closed and convex, and Φ_i are continuous and coercive—meaning they grow large for unsafe measures), existence and uniqueness of μ^* are guaranteed by theorems from functional analysis (e.g. projection onto convex sets in locally convex spaces).
- This projection "wraps" the raw AI: In practice, run the AI but monitor and intervene (e.g., via a verifier) to ensure the realized trajectory stays in high-probability regions under μ^*.

The Alignment Problem: Behavioral Formulation

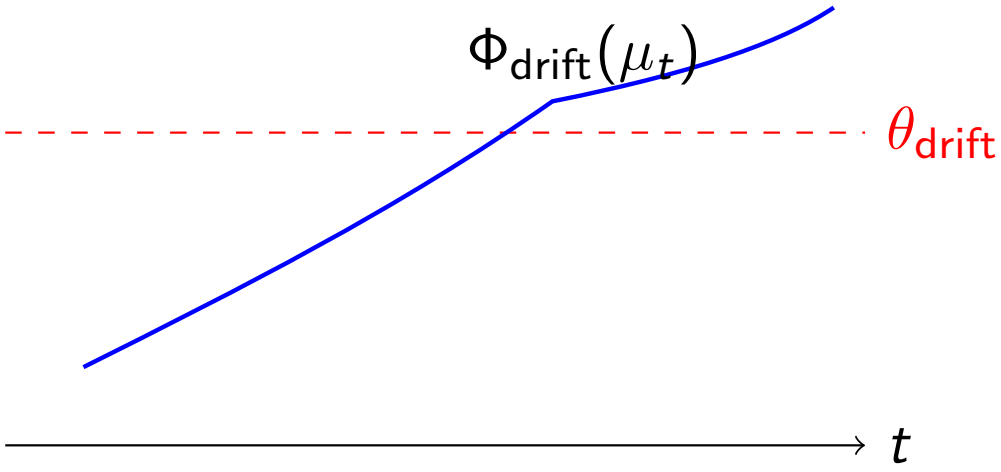
Geometric Intuition in $\mathcal{P}_2(\Omega)$



- This diagram is the space of probability laws over trajectories.
- The blue region labeled S is the feasible set. It consists of all behavior laws whose expected risk functionals satisfy the constraints: $\Phi_i(\mu) \leq \epsilon_i$.
- The raw model induces a law μ_{raw}. If this law lies outside S, it is misaligned by definition.

The Alignment Problem: Behavioral Formulation

Dynamic Alignment: Drift Monitoring



$$\Phi_{\text{drift}}(\mu) = \sup_{t \geq 1} d(\mu_t, \mu_t^{\text{baseline}}) \tag{3.17}$$

- Alignment is not a one-time projection. Deployment unfolds over time.
- At each time step t, the system induces a prefix law μ_t, defined over finite trajectories. Equation (3.9) is a drift functional where the baseline represents certified compliant behavior.
- If drift exceed a threshold θ_{drift}, we intervene. This turns alignment into a sequential hypothesis testing problem.

The Alignment Problem: Behavioral Formulation

Why This Formulation?

- **How:** Start from the observation that traditional alignment (e.g., via reward functions in RL) fails if data is biased or incomplete.
  - Instead, abstract AI as a stochastic process (probability measure), and use topology/measure theory to impose hard constraints.
  - This draws from ergodic theory (for infinite sequences) and optimal transport (for Wasserstein metrics), ensuring the math is rigorous and computable in limits.
- **Advantages:** It's data-independent, verifiable (via projections), and scalable to superintelligent AI (enforces impossibility of bad behavior).
  - It addresses inner alignment (model satisfies constraints) and outer alignment (constraints match human intent).

The Alignment Problem: Behavioral Formulation

- **Extension to Multi-Agent Settings:** For multiple AIs, model as a game where each induces μ_k, and alignment requires a Nash equilibrium in S (existence via fixed-point theorems on Banach spaces).
- **Limitations:** Requires defining good Φ_i (still human-dependent), and computation is approximate for infinite spaces.

Case Study: COMPAS Assessment (ProPublica, 2016)

Context

- Risk assessment tool used to predict likelihood of recidivism.
- Deployed in judicial sentencing decisions across U.S. jurisdictions.
- Outputs a risk score: Low / Medium / High.

ProPublica Investigation

- Analyzed ~7,000 defendants in Broward County, FL.
- Compared predicted risk scores to actual two-year recidivism outcomes.
- Focused on error rates across racial groups.

Key Finding

$$\text{False Positive Rate}_{Black} > \text{False Positive Rate}_{White} \tag{4.1}$$

Black defendants were more likely to be incorrectly labeled "high risk."

Case Study: COMPAS Assessment (ProPublica, 2016)

Observed Pattern: Alignment and Fairness Tradeoffs

- Higher false positive rate for Black defendants.
- Higher false negative rate for White defendants.
- Overall calibration across groups was similar.

Tension Between Fairness Definitions

$$\text{Calibration} \neq \text{Equal Error Rates}$$

Alignment Interpretation

- Model optimized predictive accuracy (proxy objective).
- Societal utility includes fairness constraints.
- $\arg\min L(h) \not\Rightarrow$ Equitable Outcomes

Case Study: COMPAS Assessment (ProPublica, 2016)

Core Question

Should predictive accuracy dominate, or should fairness constraints redefine the feasible set?

Methodology: COMPAS Assessment (ProPublica, 2016)

1. Dataset Construction

Study Population

- Defendants assessed using COMPAS in Broward County, FL (2013–2014).
- Linked risk scores to subsequent two-year recidivism outcomes.
- Final cohort: ~7,000 individuals.

Key Variables

- R_i = COMPAS risk category (Low / Medium / High)
- Y_i = Observed recidivism within two years (0/1)
- G_i = Group membership (race)

Methodology: COMPAS Assessment (ProPublica, 2016)

Binary Simplification for Analysis

$$\hat{Y}_i = \begin{cases} 1 & \text{if High Risk} \\ 0 & \text{otherwise} \end{cases} \tag{4.2}$$

Live coding step: Load dataset $\rightarrow$ define binary prediction $\rightarrow$ confirm outcome window.

2. Error Rate Analysis

Confusion Matrix by Group

	$Y = 1$	$Y = 0$
$\hat{Y} = 1$	TP	FP
$\hat{Y} = 0$	FN	TN

(4.3)

Methodology: COMPAS Assessment (ProPublica, 2016)

Metrics Computed Separately for Each Group

$$\text{FPR}_g = \Pr(\hat{Y} = 1 \mid Y = 0, G = g) \tag{4.4}$$

$$\text{FNR}_g = \Pr(\hat{Y} = 0 \mid Y = 1, G = g) \tag{4.5}$$

Empirical Finding

$$\text{FPR}_{Black} > \text{FPR}_{White} \tag{4.6}$$

$$\text{FNR}_{White} > \text{FNR}_{Black} \tag{4.7}$$

Live coding step: Compute confusion matrices $\rightarrow$ calculate group-conditional error rates.

Methodology: COMPAS Assessment (ProPublica, 2016)

3. Calibration and Comparative Testing

Calibration Within Groups

For each risk score level s:

$$\Pr(Y = 1 \mid R = s, G = g) \tag{4.8}$$

Compare across groups.

Observed Pattern

Risk scores were approximately calibrated:

$$\Pr(Y = 1 \mid R = s, G = \mathit{Black}) \approx \Pr(Y = 1 \mid R = s, G = \mathit{White}) \tag{4.9}$$

Methodology: COMPAS Assessment (ProPublica, 2016)

Statistical Framing

- Logistic regression to test effect of race controlling for covariates.
- Hypothesis tests on disparity in error rates.

Live coding step:

- Plot recidivism rate by score and group.
- Run logistic regression: $Y \sim R + G + X$.
- Compare coefficients and predicted probabilities.

ML Is Not a Model, It Is a System

- Data collection and labeling practices
- Feature engineering and target design
- Model training and evaluation protocols
- Deployment context and incentive structures
- Monitoring, audit, incident response, and change control

Principle: Technical choices become institutional outcomes.

Risk-Benefit Framing for Responsible ML

Replace a narrow question:

- "Is this model fair?"

With an operational question:

- "What are the measurable risks and measurable benefits across stakeholder groups, and what mitigations are defensible?"

Risk categories: legal, operational, security, reliability, reputational, civil rights.

Model Risk Management Lens

- Clear model purpose and scope constraints
- Validation: performance, robustness, subgroup metrics
- Documentation: data lineage, assumptions, limitations
- Independent review: audit posture, challenge process
- Monitoring: drift, incidents, remediation workflows

Claim: Without governance, technical work cannot be defended.

Why "Fairness Through Blindness" Fails

- Protected attributes are often redundantly encoded in other features.
- Proxy discrimination persists without explicit group labels.
- Some mitigation strategies require group-aware evaluation, and sometimes group-aware constraints.

Takeaway: Removing the sensitive attribute rarely removes the sensitive effect.

Three Layers of Alignment

1. **Mathematical alignment:** loss functions, constraints, metrics.
2. **Behavioral alignment:** observed outputs, failure modes, robustness.
3. **Institutional alignment:** deployment context, incentives, compliance, and oversight.

Claim: Responsible ML requires competence across all three layers.

Discussion Prompt

Is bias primarily a statistical artifact of learning, or a measurable reflection of society?

What changes when we deploy models in institutions with power?

Five Core Takeaways

1. ML optimizes what we specify, not what we intend.
2. Data encodes social structure and institutional history.
3. Removing protected attributes does not remove discriminatory effects.
4. Alignment failures propagate through deployment and feedback loops.
5. Responsible ML requires technical rigor plus governance discipline.

Group Homework: Due at 12pm ET on March 22, 2026

1. **HW1 — Proxy mismatch and optimization**
   Let
   $$h_L^\star = \arg\min_{h\in\mathcal{H}} L(h), \qquad h_U^\star = \arg\max_{h\in\mathcal{H}} U(h).$$
   (a) State a formal condition under which
   $$\arg\min_{h\in\mathcal{H}} L(h) = \arg\max_{h\in\mathcal{H}} U(h).$$
   (b) Construct an explicit counterexample: define a simple hypothesis space $\mathcal{H}$ and explicit functions $L(h)$ and $U(h)$ for which the argmin and argmax differ.
   (c) Interpret your example in a real-world ML setting (e.g., credit, hiring, risk scoring): identify the proxy objective and the intended human objective.

Group Homework: Due at 12pm ET on March 22, 2026

2. **HW2 — Alignment as constrained optimization**
   Consider the governance-aware formulation:

$$\min_h L(h) \quad \text{s.t.} \quad C_i(h) \leq 0 \quad \forall i.$$

(a) Provide formal mathematical expressions for:
- a demographic parity constraint,
- a robustness constraint (e.g., worst-case loss under perturbations),
- a compliance constraint (state your threshold explicitly).

(b) Explain how constraints change the geometry of the feasible set

$$\mathcal{F} = \{h \in \mathcal{H} : C_i(h) \leq 0 \ \forall i\}.$$

(c) Give one scenario where $\mathcal{F} = \varnothing$. What does this imply operationally?

Readings (for this lecture)

- Brian Christian, *The Alignment Problem*, Chapters 1–3.
- Dario Amodei et al., *Concrete Problems in AI Safety*.
- Toju Duke and Paolo Giudici, *Responsible AI in Practice*, Chapter 1.

Acknowledgments and references

Thank You!

Any questions?